

Music Genre/Sentiment Classifier

By: Deep Learners

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Github: <https://github.com/prasadsreedevi1/Deep-Learning-Final-Project/tree/main>

1. Introduction

Music listeners often struggle to find songs that match their specific niche genre and mood preferences, and curated playlists on platforms like Spotify are not always efficient at capturing the exact sound or feeling users seek. Classification of songs on Spotify through playlists can be biased and unreliable due to human error. To address this, our project explores how Convolutional Neural Networks (CNNs) and LSTMs can be used not only for algorithmic genre classification but also for predicting emotional properties of music, specifically valence and arousal scores.

1.1. Related Work

We drew inspiration from the paper titled "An Evaluation of Convolutional Neural Networks for Music Classification Using Spectrograms" (Costa et al., 2017), which compared the performance of traditional hand-crafted features and CNNs for music genre classification across Latin, Western, and African music databases. Their study found that CNNs significantly outperformed conventional methods and offered a strong alternative for automatic music characterization. Building on these findings, we frame genre classification as a classification problem using audio features extracted from spectrograms, and predict emotional attributes as a regression problem, aiming to better capture both the categorical and emotional characteristics of music to enhance the user experience in music discovery.

2. Methodology

2.1. Data

We used the DEAM dataset, which contains 1802 excerpts and full songs, each annotated with labels of the genre as well as valence and arousal values both continuously (per-second) over the entire song. For context, valence and arousal are indicators of emotions, where valence measures on a scale of positive to negative for emotions while arousal refers to the intensity. In this paper, valence-arousal is our way of measuring emotional prediction.

Each song excerpt in the dataset was 45 seconds long, and provided in a MP3 format. The valence/arousal annotations were in csv format, containing data on average and standard deviation of arousal and valence continuous annotation for each song. Alongside, there was metadata including song title, genre, and artist. We decided on this dataset because it included labels for genre as well as valence and arousal, which was necessary for genre classification and predicting emotional properties of music. It was the only dataset we found that had both genre and emotion labels.

2.2. Preprocessing

To prepare the DEAM dataset for training, we first converted each audio file into mel spectrograms using the librosa library in Python. As recommended by the dataset authors, we excluded the first 15 seconds of each 45-second audio clip due to the instability of the start of the clips. With the remaining 30 seconds clip, we segmented it into three 10-second individual clips per song. We also made sure that segments from the same clip were either both in training or both in test, as this would prevent our model from “cheating” by seeing data from the same song during both training and testing. For each segment, we generated mel spectrograms and saved them as PNG images to serve as input to our models. This tripled the amount of training data while maintaining the meaningful temporal structure.

For the label data, we merged song-level valence and arousal annotations with genre metadata from the CSV sources. Valence and arousal scores were normalized to [0,1] range. Because the genre metadata was inconsistent, we implemented a mapping function to reduce noisy genre tags into a set of four core genres: “classical,” “electronic,” “country,” and “blues.” Due to the unbalanced nature of the dataset, where genres had unbalanced amounts of songs, we had to cut down the data from 13 genres to 4 genres.

Next, each spectrogram image was resized to 256x256 pixels and valence and arousal values normalized to the [0,1] range. To enable temporal modeling with LSTMs, we split each spectrogram along the horizontal (time) axis into non-overlapping segments of width 32 pixels. These segments were grouped into sequences of 10 time steps, and flattened into 1D feature vectors per time step, allowing us an input tensor we could use for our models. Finally, we split the dataset by song ID into an 80/20 train-test split, and saved all label files as csvs.

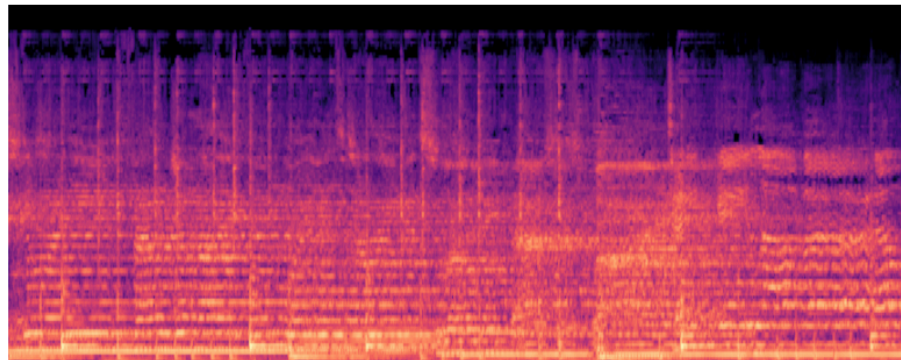


Fig 1. This is an example of a spectrogram (before resizing).

3. Model Architecture

Our implementation includes three multi-task learning models to predict both musical genre (classification) and valence-arousal (regression): a Convolutional Neural Network (CNN) model, a LSTM model, and a hybrid LSTM-CNN model. All models were designed to process 256x256 RGB spectrograms and output a genre label out of the 4 classes and a pair of continuous valence-arousal scores.

3.1. CNN Model

Our CNN model consists of four convolutional blocks with progressively increasing filter sizes (32, 64, 128, 256). Each convolutional layer is followed by batch normalization, ReLu activation, and max pooling. We apply global average pooling to convert the final feature map into a vector. To regularize training and improve generalization, we use a combination of L2 regularization, dropout with a rate of 0.5, and data augmentation techniques. Such techniques included Random Zoom, Random Translation, Contrast Variation, and horizontal flipping.

The final feature vector is passed to two heads, a softmax-activated dense layer for genre classification, and a sigmoid-activated dense layer for normalized valence-arousal regression. We chose sigmoid as sigmoid outputs values between 0 and 1, which aligns with the valence-arousal values that we also normalized between 0 and 1. The overall loss is a weighted combination of sparse cross-entropy loss for genre and mean squared error (MSE) for valence-arousal, with the regression loss scaled by a factor of 5. The motivation for using a CNN came from inspiration from the paper we referenced in addition to the fact that CNNs capture spatial relations in images, which could translate to features like tempo, dynamics, etc. for our spectrograms.

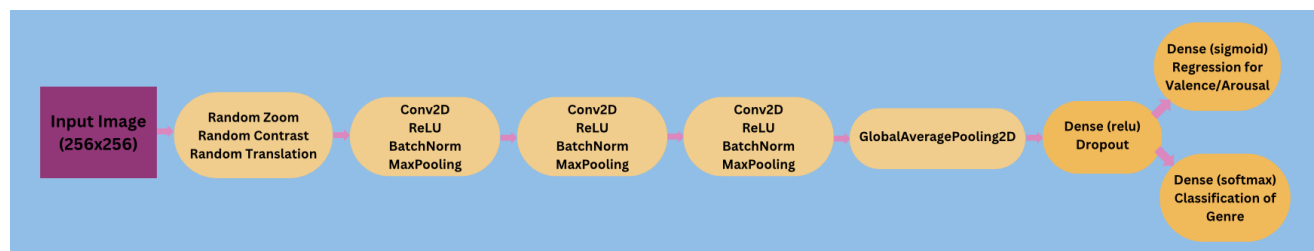


Fig 2. Convolutional Neural Network architecture. Model diagram created in Canva by us!

3.2. LSTM Model

We decided to experiment with a LSTM model because the emotion that a piece of music induces relies on the sequential structure of the sounds. We started by segmenting the spectrograms along the time (horizontal) axis in order to generate “sequential” data for the LSTM. The segmented inputs were then passed through an initial dense layer, a Bidirectional LSTM, and then another dense layer before branching into two separate heads for classification and regression.

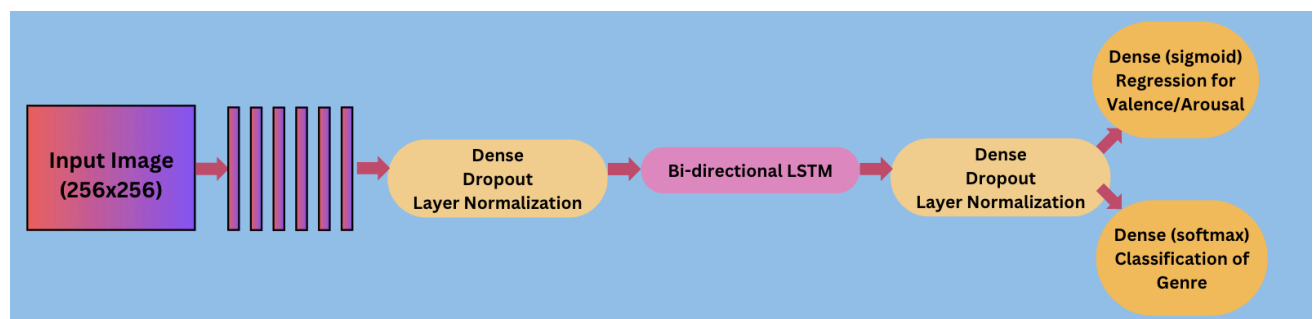


Fig 3. LSTM architecture. Model diagram created in Canva by us!

3.3. LSTM-CNN Hybrid Model

To improve our model's accuracy, we decided to experiment combining both models to create a LSTM-CNN hybrid model. In order to do this, we added a Conv2D block (i.e. Conv2D, BatchNormalization, and MaxPooling) to the inputs prior to passing them into the Bidirectional LSTM. These inputs were thin "slices" of the initial spectrogram, so the intent was that the Conv2D layer could learn spatial features that remained in these small segments. This output goes through a shared dense layer to represent the common representation for both the classification and regression tasks.

This shared representation then branches into two heads, the genre and emotion prediction head. The genre classification head is followed by the softmax output layer to classify the genre while the emotion prediction head is followed by the sigmoid output layer to predict the valence-arousal scores.

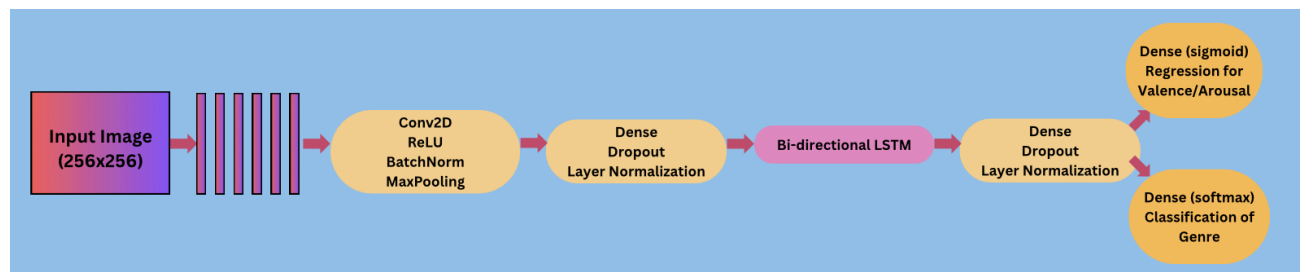


Fig 4. LSTM-CNN hybrid architecture. Model diagram created in Canva by us!

The total loss for genre classification was the sum of sparse categorical cross-entropy and for valence-arousal, the mean squared error with the regression loss scaled by a factor of 5. Our primary metrics were accuracy for genre classification and mean absolute error (MAE) for the emotion prediction. For all three models, we trained for around 75-100 epochs with an Adam optimizer (1e-3), reduceLROnPlateau, and a batch size of 64.

4. Results

GENRE CLASSIFICATION: CNN CONFUSION MATRIX

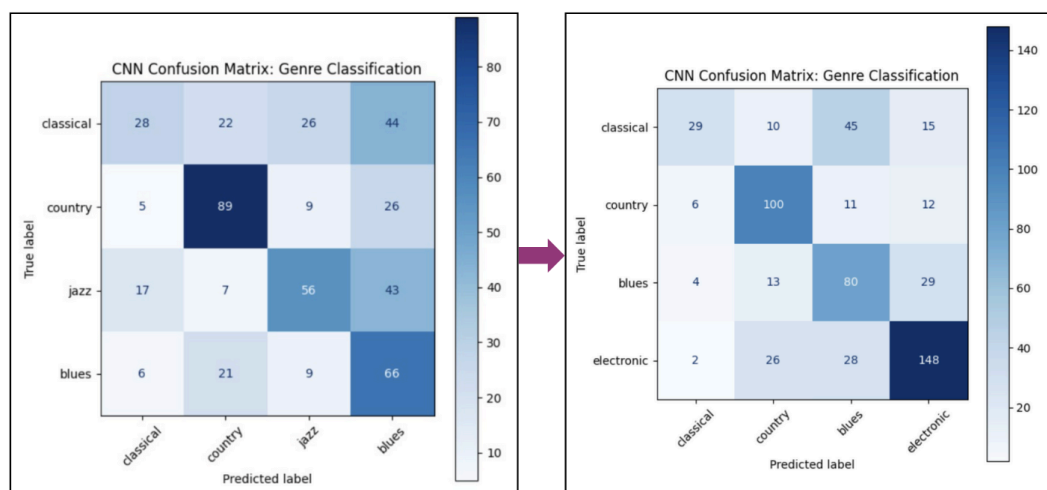


Fig 5. For the CNN model, this is the confusion matrix comparing true and predicted labels of genres before and after changing genres. The left confusion matrix was with “classical”, “country”, “jazz”, and “blues”. Because of the model’s misclassification of “classical” and “jazz,” we replaced “jazz” with “electronic,” as seen by the confusion matrix to the right.

When we first generated a confusion matrix comparing true and predicted labels of genres, we discovered that the CNN model classified “country” and “jazz” reliably, but frequently confused “classical” with “jazz” and “blues”. This suggests that certain genres in our dataset had overlapping acoustic features or annotation inconsistencies, which may have led to misclassifications. In response to this, we decided to replace “jazz” with “electronic,” which helped increase our accuracy for genre classification.

GENRE CLASSIFICATION: LSTM & LSTM-CNN CONFUSION MATRIX

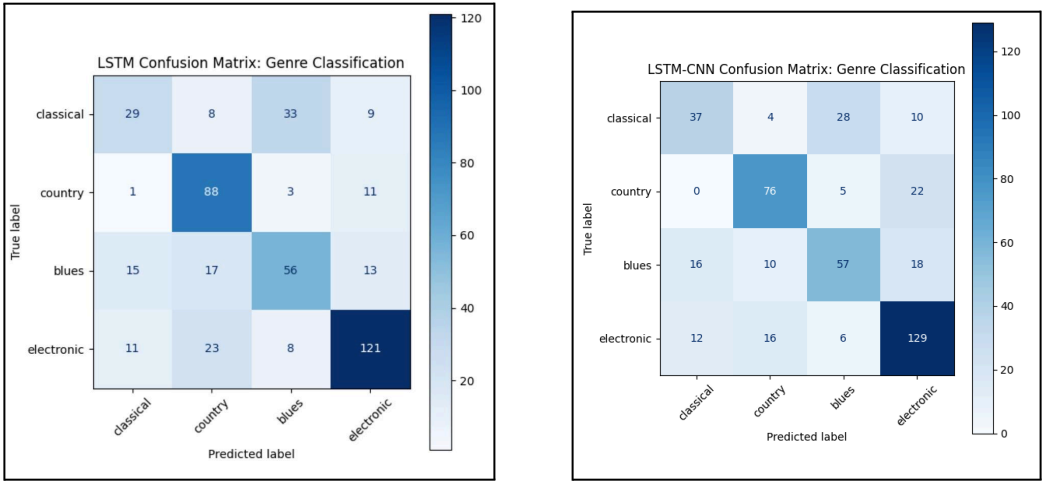


Fig 6 (LEFT). For the LSTM model, this shows the confusion matrix comparing true and predicted labels of genres.
Fig 7 (RIGHT). For the LSTM-CNN model, this shows the confusion matrix comparing true and predicted labels of genres.

The LSTM captured some temporal patterns, improving “classical” predictions slightly but performed worse on “blues,” likely due to limited spatial awareness. The LSTM-CNN hybrid provided the most balanced results, achieving better accuracy on “classical” and “blues” while still performing well on “electronic” and “country.” This suggests that combining temporal modeling with spatial feature extraction enables the model to generalize better across genres with varying structural and timbral characteristics.

GENRE CLASSIFICATION: ACCURACY + LOSS

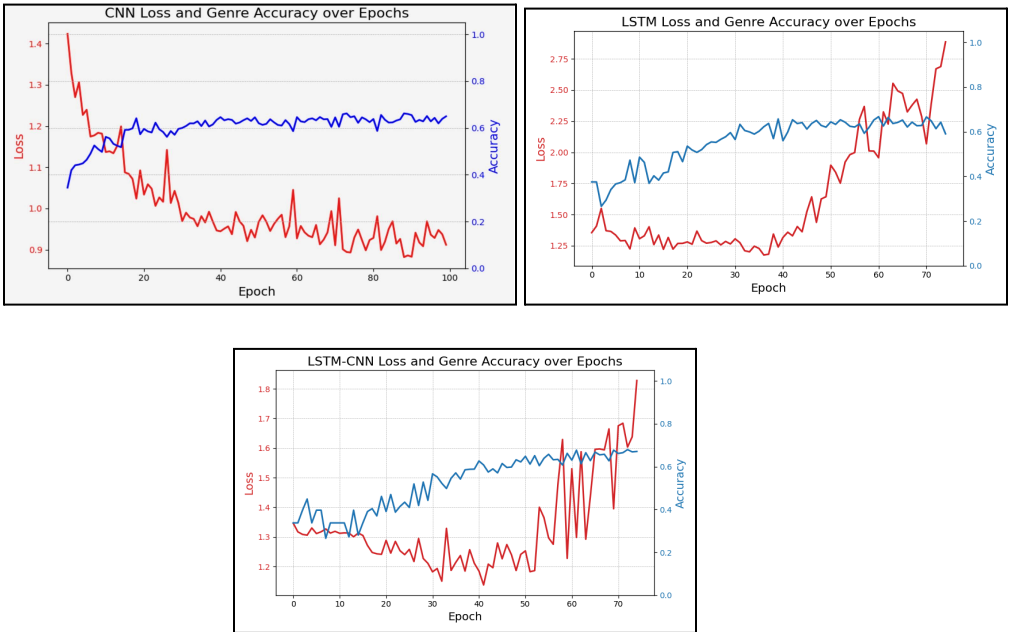


Fig 8. These graphs display the accuracy and loss for genre classification across epochs for each of the three models.

EMOTION PREDICTION: MEAN ABSOLUTE ERROR (MAE)

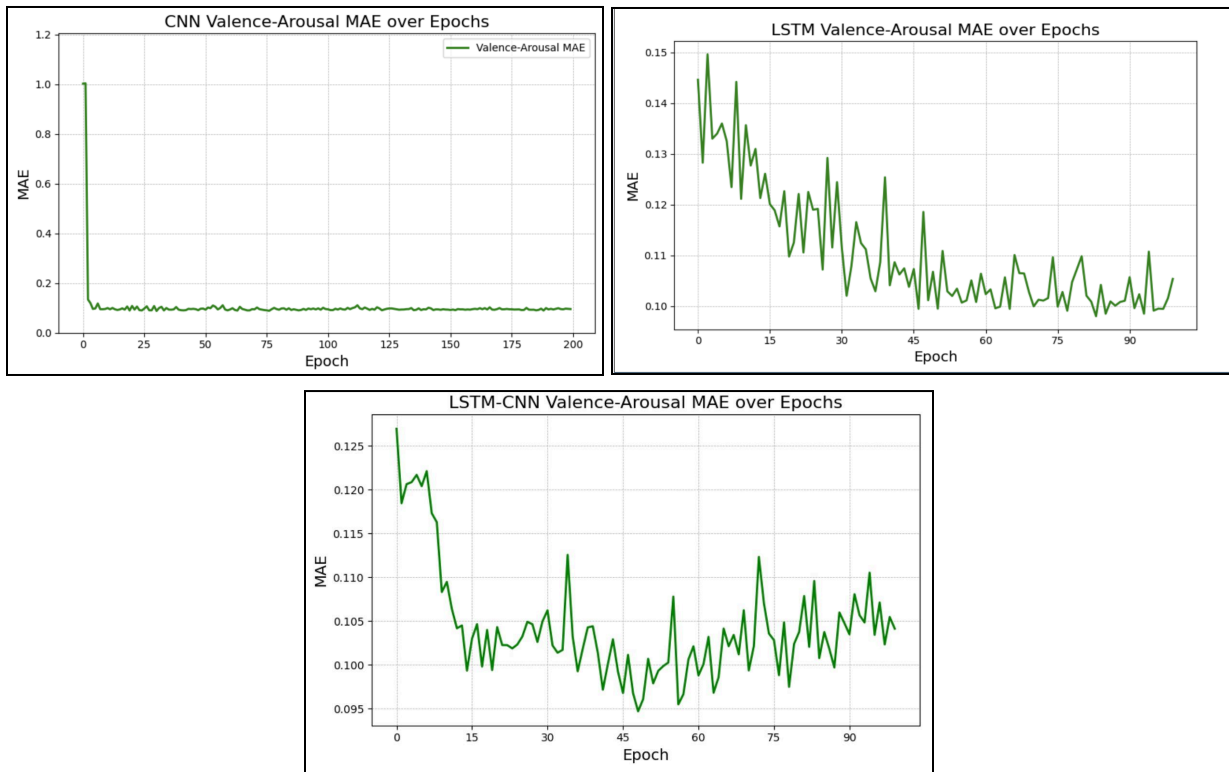


Fig 9. These graphs display the mean absolute error for emotion prediction across epochs for each of the three models.

QUANTITATIVE RESULTS

Over several trials, these are our best results we achieved for our models:

Model	Accuracy (Genre Classification)	Mean Absolute Error (Emotion Prediction)
CNN	67%	.088
LSTM	67%	.110
LSTM + CNN	69%	.090

Fig 10. Accuracy for Genre Classification and MAE for Emotion Prediction for CNN, LSTM, and LSTM+CNN.

Ultimately, we found that the CNN and LSTM-CNN hybrid performed best for emotion prediction while all models achieved relatively similar accuracy for genre classification. These results align with some of our hypotheses.

In regards to genre classification, all models achieved similar accuracy for genre classification (67-69%). CNN models can learn spatial relations and visible features in the spectrograms, which likely correspond to rhythm, timbre, volume, instrumentation, and other features, which are strong predictors of genre. Even a pure LSTM can classify genre reasonably well, as short musical segments and their sequential relationships often contain enough information to identify genre. The hybrid model performed best (69% as opposed to 67% for the two individual models), which might suggest that a combination of spatial and temporal features are best for genre classification.

We also noticed that for both the LSTM and CNN-LSTM hybrid, the loss started to increase after about 40 epochs, which seems to indicate overfitting. Despite this behavior, the test accuracy continued to increase marginally. We believe that these results might be due to exploding or vanishing gradients, since the strange behavior is limited to the models that have an LSTM in them.

For emotion prediction, our best-performing models were CNN and CNN-LSTM hybrid, where they achieved a MAE of .088-0.09, which is comparable to what researchers have been able to accomplish. The CNN model achieved the lowest MAE, suggesting that spatial patterns in spectrograms (such as timbre, instrumentation, and more) may be sufficient for emotion prediction. For the LSTM model, it displayed a gradual and slow improvement of performance and achieved a slightly higher MAE of 0.110. This could be because the LSTM captures temporal relationships, which are essential to how emotion is conveyed in music. Emotions are built from the progression and sequence of notes and chords over time. However, the LSTM may struggle to model long-range dependencies across the entire audio clip, limiting its ability to fully capture the emotions of a song.

5. Challenges

The hardest part of the project was improving our model's test accuracy due to limitations of the dataset. Initially, we were stuck at about ~20-30% test accuracy, despite multiple training attempts. This forced us to rethink our data preprocessing and model architecture, and we started with adding more convolutional layers, allowing the model to capture more complex features from the input data. Additionally, we had the idea to augment our dataset, since we thought having more training data would enable the model to perform better. So, for each 45 second clip, we experimented with segmenting it into smaller clips and keeping the same labels. Thus, we could increase the amount of data we had by a factor of the number of segments we decided to use. We segmented each audio file into 3 segments (skipped the first 15 seconds, and then the remaining 30 seconds were split into 10 second segments). This essentially increased the data we had by a factor of 3 and significantly improved accuracy.

In addition, the dataset was pretty unbalanced, as some genres did not have as many songs as others. Since our dataset involved 13 genres and it was unbalanced, we decided to preprocess our data to be more balanced and picked the 4 most represented genres, which helped improve our model accuracy by 30%. The unbalanced nature of the dataset meant that the model didn't have enough data to learn certain genres, and those underrepresented genres were essentially just "confusing" the model and hindering its

ability to learn unique features for other genres. These key changes allowed us to pass the low testing threshold and significantly improved our testing accuracy.

Another issue was labeling in the dataset. While the valence and arousal scores were very well labeled for songs, we found that the metadata for the songs that included genre wasn't maintained well. Instead of having clean, single, labels, many songs were labeled as "SoulRB-Pop-Rock" and one csv file containing a few songs were corrupted. However, this was the only dataset we found that had both genres and emotion labels. We believe that this issue was a major factor in hitting a ceiling for accuracy in genre classification in our project.

6. Discussion

When creating any kind of model, one must consider the implications of the model when implemented in the real world. One limitation of our model is that it classifies music into 4 discrete genres. In the real world, this could have negative implications for several stakeholders. For example, artists who create music that cannot neatly be distinguished into a single genre would struggle to have their music accurately classified. Additionally, we found that the DEAM dataset contained vague genre descriptions such as "International," which doesn't account for the vast variety of international music that exists. Ultimately, when making a model that classifies on genre, it is essential that the dataset is truly representative of *all* music that it intends to classify.

In the future, we would try implementing an attention-based CNN or transformer model. These options would be alternatives to LSTMs that still enable us to incorporate the temporal nature of music in our model. The attention-based CNN could potentially learn what specific parts of a music segment are most predictive of emotion. Another idea we had was to try creating our own dataset by scraping Apple Music or Spotify. This could help to mitigate some of the challenges we faced, particularly in terms of the unbalanced amount of samples for each genre and unreliable labeling. We could also try incorporating other datasets, such as the GTZAN dataset, for genre classification.

7. Reflection

Ultimately, we are proud of how our project turned out. While we didn't meet our model accuracy expectations for genre classification, we exceeded our own expectations with emotion labeling and felt as though we did the best we could given the limitations present. We successfully met our base goal by reimplementing a CNN-based classifier for genre classification. Our initial target goal was to 1) build a LSTM model for emotion prediction, and 2) build a model that predicts both genre and emotion. We went beyond these goals, as we 1) revised our CNN model to not only classify genre, but also to predict emotion, 2) built an LSTM-only model to not only predict emotion, but also to classify genre, and 3) built a hybrid LSTM-CNN model to classify both genre and predict emotion. We also somewhat met our stretch goal by hypertuning our parameters to get the best parameters we could. Although genre classification proved more difficult than we anticipated, we were still able to build models that learned meaningful patterns. For emotion prediction, our MAE is comparable to what researchers have been able to accomplish.

Over the course of the project, our approach evolved significantly. We initially intended to use a CNN for genre and an LSTM for emotion, but we decided to experiment with a hybrid model that combined CNNs for spatial feature extraction and LSTMs for temporal modeling. If we could do our project over again, we might consider building our own dataset or curating a cleaner one, if feasible. Given more time, we would also explore more innovative preprocessing strategies and potentially experiment with attention mechanisms or transformers. In addition, we could have experimented with making our model more complex, such as predicting folksonomy tags, or multi-label attributes for the songs.

This project exposed us to the struggles that researchers face in real life: uncertain data, choosing appropriate model architectures, resource limits, and the overall black-box that training neural network models can be. Our biggest takeaway is that data in the real world is not always going to be perfect and that as researchers, we have to think outside the box to deal with these challenges.